

Lecture 18: Exponentially Affine Models in Discrete Time

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Intro

- Last lecture: EH failed, forwards forecast returns, Diebold-Li summarized the curve
- Today: build a no-arbitrage model that separates expected rates from term premia
- Goal: move from reduced-form curve dynamics to a structural pricing system
- This is where $\mathbb{P}$, $\mathbb{Q}$, and λ_t start doing real work

Why EH Is Not Enough

- EH was a useful benchmark, but too restrictive
- It ruled out time-varying expected bond returns
- The data did not cooperate ([Fama and Bliss 1987](#); [Campbell and Shiller 1991](#))
- We need a model where the whole curve is priced jointly
- And where expected short rates and term premia are different objects

What This Model Will Deliver

- A state vector X_t
- Dynamics under the physical measure $\mathbb{P}$
- An explicit market price of risk λ_t
- Risk-neutral dynamics under $\mathbb{Q}$
- Bond prices, yields, risk-neutral yields, and term premia

- Part I: affine setup under $\mathbb{P}$
- Part II: change of measure and why pricing uses $\mathbb{Q}$
- Part III: exponential-affine bond pricing
- Part IV: risk-neutral yields and term premia
- Part V: implementation and estimation with observed factors only

Part I: Setup Under $\mathbb{P}$

- Let $X_t \in \mathbb{R}^K$ collect the term-structure factors
- In this lecture, X_t is *observed*
- Think: first few PCs of yields, or a small set of yields

$$X_{t+1} = \mu + \Phi X_t + \Sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, I_K)$$

- $\mathbb{P}$ is the actual law of motion in the data

$$y_t^{(1)} = \delta_0 + \delta_1^\top X_t$$

- Same notation as last lecture
- The one-period yield is the short rate
- The short end of the curve moves with the state
- Once X_t is given, the short rate is pinned down

$$m_{t+1} = -y_t^{(1)} - \frac{1}{2}\lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}$$

$$\lambda_t = \lambda_0 + \lambda_1 X_t$$

- The SDF comes first: it is the pricing object
- λ_t is the market price of risk embedded inside it
- If λ_t changes with X_t , bond premia change with the state
- No Λ_1 : we use λ_1

- $\mathbb{P}$: how the state actually evolves
- m_{t+1} : how payoffs are discounted
- λ_t : how much compensation investors require for bond risk
- If λ_t varies with X_t , term premia vary with X_t
- That is already something EH could not do

Questions?

Part II: Change of Measure

Define

$$\frac{d\mathbb{Q}}{d\mathbb{P}}\Big|_{t+1} = \exp\left(-\frac{1}{2}\lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}\right)$$

- This is the one-step Radon-Nikodym derivative
- It tilts probabilities state by state
- We now need to check that this is a valid density
- Notice that it is a positive random variable

Quick Recall: Gaussian MGF

If $\varepsilon_{t+1} \sim N(0, I_K)$ under $\mathbb{P}$, then for any a_t known at time t ,

$$\mathbb{E}_t^{\mathbb{P}} \left[e^{a_t^\top \varepsilon_{t+1}} \right] = \exp \left(\frac{1}{2} a_t^\top a_t \right)$$

- We will use this twice
- First to show the density integrates to one
- Then to prove the shifted shock is standard normal under $\mathbb{Q}$

It Integrates to One

Conditioning on time t ,

$$\mathbb{E}_t^{\mathbb{P}} \left[\frac{dQ}{dP} \Big|_{t+1} \right] = e^{-\frac{1}{2} \lambda_t^\top \lambda_t} \mathbb{E}_t^{\mathbb{P}} \left[e^{-\lambda_t^\top \varepsilon_{t+1}} \right]$$

Applying the Gaussian mgf with $a_t = -\lambda_t$,

$$\mathbb{E}_t^{\mathbb{P}} \left[e^{-\lambda_t^\top \varepsilon_{t+1}} \right] = \exp \left(\frac{1}{2} \lambda_t^\top \lambda_t \right)$$

so

$$\mathbb{E}_t^{\mathbb{P}} \left[\frac{dQ}{dP} \Big|_{t+1} \right] = 1$$

Define

$$\varepsilon_{t+1}^{\mathbb{Q}} \equiv \varepsilon_{t+1} + \lambda_t$$

- Under $\mathbb{Q}$, this will become the convenient innovation
- Same state variables, different conditional law
- We now prove explicitly that $\varepsilon_{t+1}^{\mathbb{Q}} \sim N(0, I_K)$ under $\mathbb{Q}$

For any $u \in \mathbb{R}^K$,

$$\begin{aligned}\mathbb{E}_t^{\mathbb{Q}} \left[e^{u^\top \varepsilon_{t+1}^{\mathbb{Q}}} \right] &= \mathbb{E}_t^{\mathbb{P}} \left[\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{t+1} e^{u^\top (\varepsilon_{t+1} + \lambda_t)} \right] \\&= \exp \left(u^\top \lambda_t - \frac{1}{2} \lambda_t^\top \lambda_t \right) \mathbb{E}_t^{\mathbb{P}} \left[e^{(u - \lambda_t)^\top \varepsilon_{t+1}} \right] \\&= \exp \left(u^\top \lambda_t - \frac{1}{2} \lambda_t^\top \lambda_t + \frac{1}{2} (u - \lambda_t)^\top (u - \lambda_t) \right) \\&= \exp \left(\frac{1}{2} u^\top u \right)\end{aligned}$$

- This is the mgf of $N(0, I_K)$
- So $\varepsilon_{t+1}^{\mathbb{Q}} \sim N(0, I_K)$ under $\mathbb{Q}$

Start from the $\mathbb{P}$ -VAR and replace $\varepsilon_{t+1} = \varepsilon_{t+1}^{\mathbb{Q}} - \lambda_t$:

$$X_{t+1} = \mu + \Phi X_t + \Sigma(\varepsilon_{t+1}^{\mathbb{Q}} - \lambda_t)$$

Using $\lambda_t = \lambda_0 + \lambda_1 X_t$,

$$X_{t+1} = \underbrace{(\mu - \Sigma\lambda_0)}_{\equiv \mu^{\mathbb{Q}}} + \underbrace{(\Phi - \Sigma\lambda_1)}_{\equiv \Phi^{\mathbb{Q}}} X_t + \Sigma\varepsilon_{t+1}^{\mathbb{Q}}$$

$$P_t^{(n+1)} = \mathbb{E}_t^{\mathbb{P}}[M_{t+1}P_{t+1}^{(n)}] = e^{-y_t^{(1)}} \mathbb{E}_t^{\mathbb{Q}}[P_{t+1}^{(n)}]$$

- There is a state evolution under both $\mathbb{P}$ and $\mathbb{Q}$
- But the bond pricing recursion uses $\mathbb{Q}$
- $\mathbb{P}$ matters for forecasts and expected excess returns
- $\mathbb{Q}$ matters for prices
- This is not “just different initial conditions”: the dynamics themselves change

Questions?

Part III: Bond Pricing

For a zero-coupon bond,

$$P_t^{(n+1)} = \mathbb{E}_t^{\mathbb{P}}[M_{t+1}P_{t+1}^{(n)}] = e^{-y_t^{(1)}} \mathbb{E}_t^{\mathbb{Q}}[P_{t+1}^{(n)}]$$

- Last lecture gave the bond math
- Now we impose no arbitrage
- Once we know the short rate and the $\mathbb{Q}$ -dynamics, all bond prices are pinned down

Guess ([Duffie and Kan 1996](#))

$$P_t^{(n)} = \exp(A_n + B_n^\top X_t)$$

with

$$A_0 = 0, \quad B_0 = 0$$

- $P_t^{(0)} = 1$ is the base case
- The question is whether this functional form closes under the recursion

Plug the Guess Into the Pricing Equation

Use

$$y_t^{(1)} = \delta_0 + \delta_1^\top X_t$$

and

$$X_{t+1} = \mu^{\mathbb{Q}} + \Phi^{\mathbb{Q}} X_t + \Sigma \varepsilon_{t+1}^{\mathbb{Q}}$$

Then

$$P_t^{(n+1)} = e^{-\delta_0 - \delta_1^\top X_t} \mathbb{E}_t^{\mathbb{Q}} [\exp(A_n + B_n^\top X_{t+1})]$$

Compute the Gaussian Expectation

Substitute for X_{t+1} and collect what is known at time t :

$$P_t^{(n+1)} = e^{-\delta_0 - \delta_1^\top X_t} \mathbb{E}_t^{\mathbb{Q}} \left[\exp \left(A_n + B_n^\top (\mu^{\mathbb{Q}} + \Phi^{\mathbb{Q}} X_t + \Sigma \varepsilon_{t+1}^{\mathbb{Q}}) \right) \right]$$

$$P_t^{(n+1)} = \exp \left(A_n - \delta_0 + B_n^\top \mu^{\mathbb{Q}} + [(\Phi^{\mathbb{Q}})^\top B_n - \delta_1]^\top X_t \right) \mathbb{E}_t^{\mathbb{Q}} \left[e^{B_n^\top \Sigma \varepsilon_{t+1}^{\mathbb{Q}}} \right]$$

Because $\varepsilon_{t+1}^{\mathbb{Q}} \sim N(0, I_K)$,

$$\mathbb{E}_t^{\mathbb{Q}} \left[e^{B_n^\top \Sigma \varepsilon_{t+1}^{\mathbb{Q}}} \right] = \exp \left(\frac{1}{2} B_n^\top \Sigma \Sigma^\top B_n \right)$$

Matching coefficients gives

$$A_{n+1} = A_n + B_n^\top \mu^{\mathbb{Q}} + \frac{1}{2} B_n^\top \Sigma \Sigma^\top B_n - \delta_0$$

$$B_{n+1} = (\Phi^{\mathbb{Q}})^\top B_n - \delta_1$$

- The recursion closes
- This is why the model is called exponentially affine

Yields Are Affine Too

Using

$$y_t^{(n)} = -\frac{1}{n} \log P_t^{(n)}$$

we get

$$y_t^{(n)} = -\frac{1}{n} (A_n + B_n^\top X_t)$$

- Each maturity loads linearly on the same state vector
- The loadings vary with n through A_n and B_n
- The whole curve is one joint pricing object

- Diebold-Li also used a small number of factors to describe the curve ([Diebold and Li 2006](#))
- But there the loadings were chosen for fit and forecasting
- Here the loadings come from no-arbitrage restrictions ([Duffee 2002](#); [Ang and Piazzesi 2003](#))
- Most important: the model gives a disciplined $\mathbb{P}/\mathbb{Q}$ split

Questions?

Part IV: EH Component and Term Premia

Lecture 17 defined the expectations-hypothesis piece as

$$EH_t^{(n)} \equiv \frac{1}{n} \sum_{j=0}^{n-1} \mathbb{E}_t^{\mathbb{P}} \left[y_{t+j}^{(1)} \right]$$

- This is the average expected future short rate
- It is a $\mathbb{P}$ -expectations object
- It is *not* computed from the bond-pricing recursion
- Here we show how to compute it once the affine model is estimated

Start from

$$X_{t+1} = \mu + \Phi X_t + \Sigma \varepsilon_{t+1}, \quad \mathbb{E}_t^{\mathbb{P}}[\varepsilon_{t+1}] = 0$$

Then

$$\mathbb{E}_t^{\mathbb{P}}[X_{t+1}] = \mu + \Phi X_t$$

and, recursively,

$$\mathbb{E}_t^{\mathbb{P}}[X_{t+j}] = \sum_{i=0}^{j-1} \Phi^i \mu + \Phi^j X_t$$

Forecast Future Short Rates

Using

$$y_t^{(1)} = \delta_0 + \delta_1^\top X_t$$

we get

$$\mathbb{E}_t^\mathbb{P} [y_{t+j}^{(1)}] = \delta_0 + \delta_1^\top \mathbb{E}_t^\mathbb{P} [X_{t+j}]$$

So once $(\mu, \Phi, \delta_0, \delta_1)$ and X_t are known,

$$\mathbb{E}_t^\mathbb{P} [y_{t+j}^{(1)}] = \delta_0 + \delta_1^\top \left(\sum_{i=0}^{j-1} \Phi^i \mu + \Phi^j X_t \right)$$

Compute the EH Component

Now average those expected short rates:

$$EH_t^{(n)} = \frac{1}{n} \sum_{j=0}^{n-1} \mathbb{E}_t^{\mathbb{P}} [y_{t+j}^{(1)}]$$

Algorithm:

1. Start from the estimated state X_t
2. Forecast $X_{t+1}, \dots, X_{t+n-1}$ under the (model-implied) $\mathbb{P}$
3. Map each state forecast into an expected short rate
4. Average across the n horizons

Observed, Fitted, and EH Component

- Observed yield: $y_t^{(n)}$
- Fitted/model-implied yield: $\hat{y}_t^{(n)}$
- EH component: $EH_t^{(n)}$
- Term premium: $TP_t^{(n)}$

Observed, Fitted, and EH Component

- Observed yield: $y_t^{(n)}$
- Fitted/model-implied yield: $\hat{y}_t^{(n)}$
- EH component: $EH_t^{(n)}$
- Term premium: $TP_t^{(n)}$
- In an exact-fit model, $y_t^{(n)}$ and $\hat{y}_t^{(n)}$ are the same
- In an estimated model with measurement error, they need not be
- But $EH_t^{(n)}$ is different from both: it uses only $\mathbb{P}$ -expected short rates

Term Premia Relative to the EH Component

In the spirit of Lecture 17, define

$$TP_t^{(n)} = y_t^{(n)} - EH_t^{(n)}$$

or, in an inexact-fit model,

$$TP_t^{(n)} = \hat{y}_t^{(n)} - EH_t^{(n)}$$

- This is the part of the yield not attributed to expected future short rates
- If $TP_t^{(n)}$ rises, yields rise relative to the EH component
- If $TP_t^{(n)}$ falls, yields move closer to the EH component
- This creates a gap between *observed* yields and *expected* short rates

Why λ_t Still Matters

- The EH component uses $\mathbb{P}$ forecasts of future short rates
- The fitted yield comes from the no-arbitrage pricing recursion
- The wedge between $\mathbb{P}$ and $\mathbb{Q}$ is driven by $\lambda_t = \lambda_0 + \lambda_1 X_t$
- So λ_t is still the key force behind time-varying term premia

Questions?

Part V: Implementation and Estimation

Inputs After Estimation

Suppose we have estimated

$$\mu, \Phi, \Sigma, \delta_0, \delta_1, \lambda_0, \lambda_1$$

Then compute

$$\mu^{\mathbb{Q}} = \mu - \Sigma\lambda_0, \quad \Phi^{\mathbb{Q}} = \Phi - \Sigma\lambda_1$$

- The fitted yield curve still uses the pricing recursion with $\mu^{\mathbb{Q}}, \Phi^{\mathbb{Q}}$
- The EH component uses the $\mathbb{P}$ -dynamics (μ, Φ)
- After estimation, both objects are available

EH Component Algorithm

For each horizon j ,

$$\mathbb{E}_t^{\mathbb{P}}[X_{t+j}] = \sum_{i=0}^{j-1} \Phi^i \mu + \Phi^j X_t$$

Then compute the expected future short rate:

$$\mathbb{E}_t^{\mathbb{P}}[y_{t+j}^{(1)}] = \delta_0 + \delta_1^{\top} \mathbb{E}_t^{\mathbb{P}}[X_{t+j}]$$

Finally,

$$EH_t^{(n)} = \frac{1}{n} \sum_{j=0}^{n-1} \mathbb{E}_t^{\mathbb{P}}[y_{t+j}^{(1)}]$$

From Fitted Yields to Term Premia

1. Use the no-arbitrage recursion to compute fitted yields $\hat{y}_t^{(n)}$
2. Use the $\mathbb{P}$ -VAR and the short-rate equation to compute $EH_t^{(n)}$
3. Define term premia as

$$TP_t^{(n)} = \hat{y}_t^{(n)} - EH_t^{(n)}$$

In an exact-fit model, this is the same as

$$TP_t^{(n)} = y_t^{(n)} - EH_t^{(n)}$$

Questions?

Estimation Route: Observed Factors Only

1. Choose observed factors X_t
2. Estimate the VAR under $\mathbb{P}$ by OLS
3. Estimate the short-rate equation
4. Choose λ_0, λ_1 to match the yield curve
5. Recover fitted yields, the EH component, and term premia

See also Joslin et al. ([2011](#)) and Adrian et al. ([2013](#)).

A simple objective is

$$\min_{\lambda_0, \lambda_1} \sum_t \sum_{n \in \mathcal{M}} \left(y_t^{(n)} - \hat{y}_t^{(n)} \right)^2$$

- $\mathcal{M}$: maturities used in estimation
- The $\mathbb{P}$ -VAR alone is not enough
- The cross section disciplines the pricing wedge

What the Estimated Model Delivers

- Fitted yield curves
- Risk-neutral yield curves
- Term-premium estimates by maturity and over time
- A clean separation between expected-rate news and premium news
- A bridge from term-structure data to macro and finance questions

Questions?

Part VI: A Simple Estimated DTSM

A Simple DTSM with NS Factors

- Build $X_t = (\beta_{1t}, \beta_{2t}, \beta_{3t})^\top$ from a fixed- λ Nelson-Siegel fit to the LW curve
- Monthly step; annualized continuously compounded yields
- Estimate under $\mathbb{P}$:

$$X_{t+1} = \mu + \Phi X_t + \Sigma \varepsilon_{t+1}$$

- Short rate and pricing wedge:

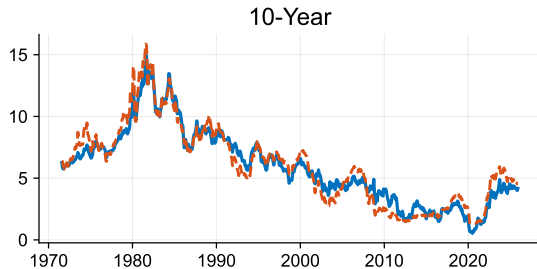
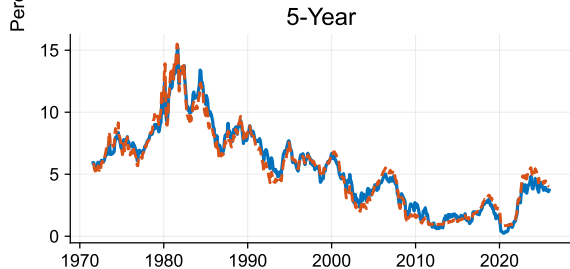
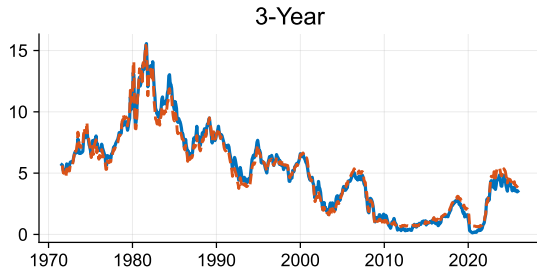
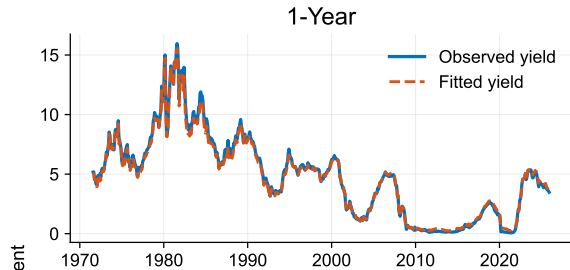
$$y_t^{(1m)} = \delta_0 + \delta_1^\top X_t, \quad \lambda_t = \lambda_0 + \lambda_1 X_t$$

- Choose (λ_0, λ_1) to fit annual maturities from 1 to 10 years

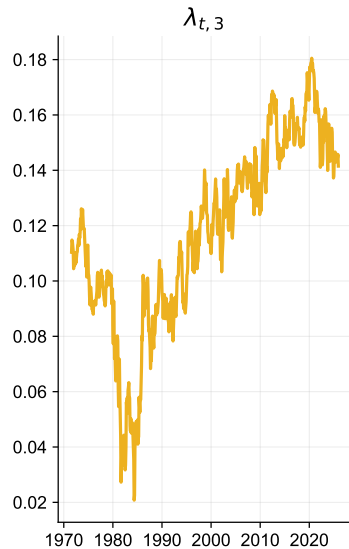
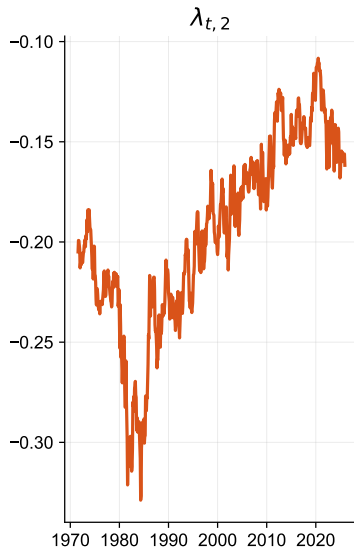
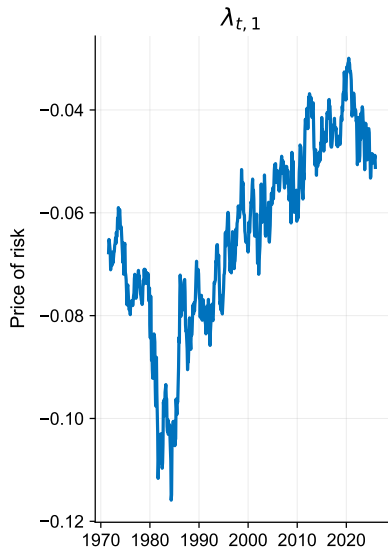
What We Estimate

- Data: LW monthly zero-coupon yields, balanced sample 1971:08–2025:12
- State vector: Nelson-Siegel level, slope, and curvature with fixed $\lambda = 0.0609$ (Diebold and Li 2006)
- $\mathbb{P}$ -dynamics: VAR(1) for the three factors
- Cross-section fit: annual maturities from 1 to 10 years
- Reported objects: observed yield, EH component, and term premium at 1y, 3y, 5y, and 10y
- Full estimation time: 0.41 seconds

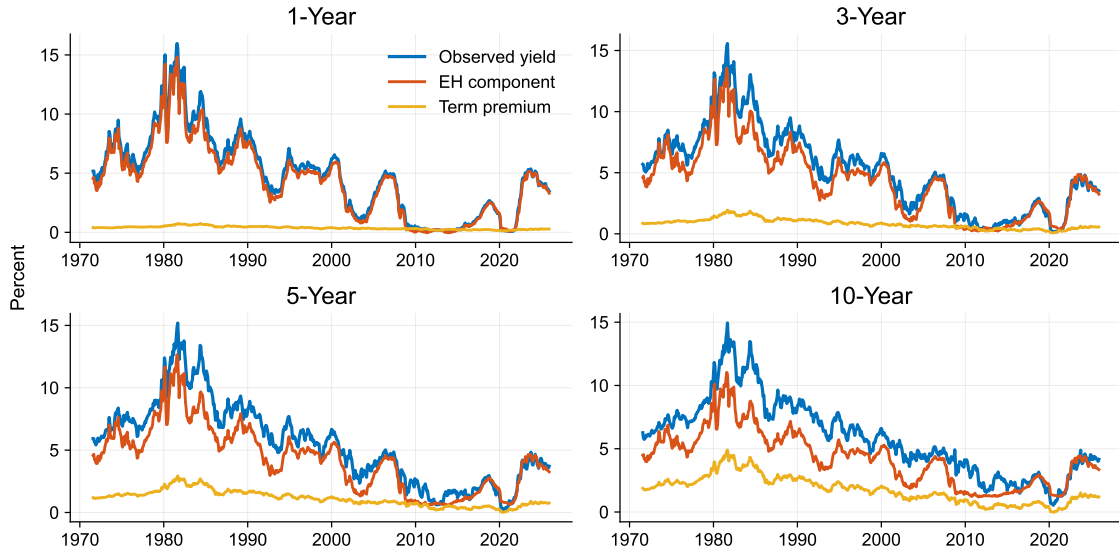
Actual and Fitted Yields



Market Prices of Risk



Yields, EH Component, and Term Premia



Questions?

- Lecture 17 told us the curve reflects both expected rates and premia
- Lecture 18 gave a no-arbitrage model that makes that separation explicit
- $\mathbb{P}$ describes the data; $\mathbb{Q}$ prices bonds
- λ_t is the wedge between the two
- Exponential-affine recursions turn that wedge into bond prices, yields, and term premia

Readings and References i

- Term-structure overview: Gürkaynak and Wright (2012)
- Evidence against EH: Fama and Bliss (1987); Campbell and Shiller (1991)
- Reduced-form benchmark: Diebold and Li (2006)
- Affine term-structure foundations: Duffie and Kan (1996); Duffee (2002); Ang and Piazzesi (2003)
- Observable-factor estimation: Joslin et al. (2011); Adrian et al. (2013)

Adrian, Tobias, Richard K. Crump, and Emanuel Moench. 2013. "Pricing the Term Structure with Linear Regressions." *Journal of Financial Economics* 110 (1): 110–38.
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